A SMOOTH SURJECTIVE COVER BY A SPACE-REPRESENTABLE STACK MAKES THE TARGET ALGEBRAIC

THE STACKS PROJECT AUTHORS

Editorial note. The source argues the 2-fibre square relation 'formally' with the two diagrams displayed; reading the conclusion off the four 2-fibre product squares is carried as an explicit obligation. The transfer inputs are not carded and are declared external.

Allowed standard prerequisites (editorial).

- Diagonal criterion. lemma-representable-diagonal (carded L26680). Reduces representability of Delta to representability of T1 x_X T2.
- 2-fibre products of stacks in setoids. Stacks, lemmas 2-fibre-product-stacks-in-setoids-over-stack-in-groupoids and stack-in-setoids-characterize (cross-chapter, named). T1 x_X T2 corresponds to a sheaf F.
- Setoid presheaf representability. lemma-map-fibred-setoids-representable-algebraic-spaces (same chapter, NOT carded, declared external). Reads representability off the sheaf correspondence.
- Base-change stability. lemma-base-change-representable-by-spaces (same chapter, NOT carded, declared external). Used in the property-transfer block.
- Setoid property transfer. lemma-map-fibred-setoids-property (same chapter, NOT carded, declared external). Transfers smooth and surjective to the sheaf map.
- Base-change property transfer. lemma-base-change-representable-transformations-property (same chapter, NOT carded, declared external). Transfers smooth and surjective along base change.
- Final bootstrap theorem. Bootstrap, theorem final-bootstrap (cross-chapter, named). Concludes F is an algebraic space.

Source reference. Original reference 04S6: focused-seeds/geometry/S9-alg-smooth-surjective-morphism-implies-algebraic/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 026O: sources/stacks-algebraic/pinned-original-source.tex, source lines 1506–1506. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:algebraic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0302: sources/stacks-algebraic/pinned-original-source.tex, source lines 876–876. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045C: sources/stacks-algebraic/pinned-original-source.tex, source lines 1180–1180. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04TC: sources/stacks-algebraic/pinned-original-source.tex, source lines 1138–1138. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04SY: sources/stacks-algebraic/pinned-original-source.tex, source lines 855–855. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 045G: sources/stacks-algebraic/pinned-original-source.tex, source lines 1337–1337. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05UJ: focused-seeds/geometry/S9-alg-smooth-surjective-morphism-implies-algebraic/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0432: focused-seeds/geometry/S9-alg-smooth-surjective-morphism-implies-algebraic/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $u : \mathcal{U} \to \mathcal{X}$ be a 1-morphism of stacks in groupoids over $(Sch/S)_{fppf}$. If*

(1) *$\mathcal{U}$ is representable by an algebraic space, and*
(2) *u is representable by algebraic spaces, surjective and smooth,*

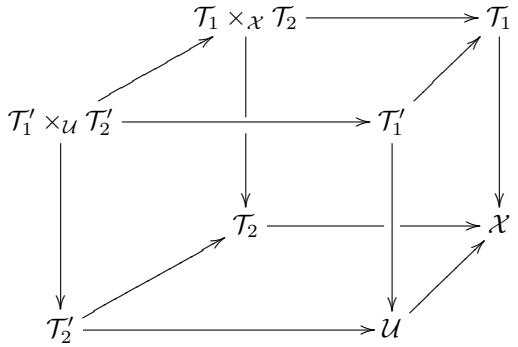
then $\mathcal{X}$ is an algebraic stack over S.

Proof. We have to show that $\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces, see Definition 026O. Given two schemes T_1, T_2 over S denote $\mathcal{T}_i = (Sch/T_i)_{fppf}$ the associated representable fibre categories. Suppose given 1-morphisms $f_i : \mathcal{T}_i \to \mathcal{X}$. According to Lemma 045G it suffices to prove that the 2-fibered product $\mathcal{T}_1 \times_{\mathcal{X}} \mathcal{T}_2$ is representable by an algebraic space. By Stacks, Lemma 05UJ this is in any case

a stack in setoids. Thus $\mathcal{T}_1 \times_{\mathcal{X}} \mathcal{T}_2$ corresponds to some sheaf F on $(Sch/S)_{fppf}$, see Stacks, Lemma 0432. Let U be the algebraic space which represents $\mathcal{U}$. By assumption

$$\mathcal{T}_i' = \mathcal{U} \times_{u,\mathcal{X},f_i} \mathcal{T}_i$$

is representable by an algebraic space T_i' over S. Hence $\mathcal{T}_1' \times_{\mathcal{U}} \mathcal{T}_2'$ is representable by the algebraic space $T_1' \times_U T_2'$. Consider the commutative diagram



In this diagram the bottom square, the right square, the back square, and the front square are 2-fibre products. A formal argument then shows that $\mathcal{T}_1' \times_{\mathcal{U}} \mathcal{T}_2' \to \mathcal{T}_1 \times_{\mathcal{X}} \mathcal{T}_2$ is the "base change" of $\mathcal{U} \to \mathcal{X}$, more precisely the diagram

$$\begin{array}{ccc} \mathcal{T}_1' \times_{\mathcal{U}} \mathcal{T}_2' & \longrightarrow & \mathcal{U} \\ \downarrow & & \downarrow \\ \mathcal{T}_1 \times_{\mathcal{X}} \mathcal{T}_2 & \longrightarrow & \mathcal{X} \end{array}$$

is a 2-fibre square. Hence $T_1' \times_U T_2' \to F$ is representable by algebraic spaces, smooth, and surjective, see Lemmas 04SY, 0302, 04TC, and 045C. Therefore F is an algebraic space by Bootstrap, Theorem 04S6 and we win. □